Lightweight RSS-Based Visible Light Positioning Considering Dust Accumulation

Ling Jin, Qinian Jiang, Shuzhou Yu, Haoyu Xu, Wenjuan E, Menxi Xie, Xiaofeng Li, *Senior Member, IEEE*, and Dong Ma

***Abstract*—The performance of received signal strength (RSS)-based visible light positioning (VLP) is heavily reliant on the pre-existing channel model which describes the RSS attenuation with distance. In indoor areas, dust particles accumulating on the surfaces of LEDs may cause the pre-existing channel model deviating from actual situations and consequently decrease the localization performance. In this paper, we first experimentally verify the impact of dust on the RSS VLP system performance and find that the positioning error is significantly increased even in slight dust deposition scenarios. Then we modify the classical Lambertian-based channel model to reflect the influence of dust, and propose a two-points approach which performs the channel model calibration only based on two RSS measurements. Additionally, the variations of the channel model parameters with the accumulation of dust are observed and mathematically modeled. A single-point approach is further designed which requires only one RSS measurement for each channel model calibration. The experimental results demonstrate that the average positioning error remains below 6 cm under different dust accumulation scenarios. The negative impact of dust can be effectively mitigated by our proposed RSS VLP scheme with low system maintenance cost. To the best of our knowledge, this is the first piece of work considering dust accumulation for VLP.**

***Index Terms*—Dust, experiment, indoor positioning system, model calibration, received signal strength (RSS), visible light positioning (VLP).**

I. Introduction

In RECENT years, emerging technologies including Internet of Things (IoT), mobile robots, and autonomous driving significantly increase the demand of indoor high-precision positioning [1], [2]. The global navigation satellite system (GNSS) has achieved great success in outdoor positioning, but cannot provide satisfied service in indoor and underground environments due to severe signal obstruction and reflection. Therefore, a variety of replaced technologies have been studied [3], [4], including WiFi, 5G, Bluetooth, radio frequency identification (RFID), ultrawideband (UWB), visible light positioning (VLP), etc. VLP systems utilize light emitting diodes (LEDs) as transmitters and thus can easily be integrated into the existing lighting infrastructure. Since VLP operates in the visible light spectrum, high speed data transmission can be achieved owing to its huge license free bandwidth. If combined with visible light communication (VLC), the VLP systems can simultaneously serve multiple purposes of illumination, communication and positioning [5], [6]. Compared to those radio frequency (RF) technologies, VLP offers several advantages including low cost, energy efficient, high security and no electromagnetic interference. In addition, multipath effects are not as significant in VLP as those in RF-based positioning because of the inherent line-of-sight (LOS) property of LEDs [7]. Centimeter-level positioning accuracy can be achieved by VLP [8]. With the widespread use of LEDs for illumination, VLP has been seen as a promising solution for indoor positioning.

In VLP systems, various methods have been proposed for localization including proximity [9], received signal strength (RSS) [2], [8], [10], [11], angle-of-arrival (AOA) [12], time-of -arrival (TOA) or time-difference-of-arrival (TDOA) [13]. Proximity is the simplest method but has limited positioning accuracy [14]. AOA can achieve good position estimation while relies on high computational complexity and expensive equipment [8]. TOA and TDOA require highly synchronized hardware which makes the system complex and impractical for implementation. RSS can be done using a single cheap photodiode (PD) and has been the most popular VLP method because of its low complexity and easy implementation [15]. VLP that utilizes RSS can either be model-based [2], [16] or fingerprinting-based [17], [18]. The model-based approach is comparatively efficient in time- and labor-consuming, and especially suitable for IoT applications [2]. Moreover, it can also be used to assist fingerprinting approaches to build high-granularity RSS fingerprint database with limited manual involvement [8], [19]. Based on these considerations, we focus on the model-based RSS VLP in this paper.

In general RSS VLP systems, a PD explores the RSS values from multiple LEDs and converts them into distances according to the pre-existing optical channel model which describes the RSS attenuation with distance. The localization can be further achieved by trilateration methods [20]. The performance of RSS VLP depends on the channel model, the RSS values and the localization methods. Existing studies mainly focus

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on improving the channel model's adaptivity as well as the localization performance, and overcoming the practical application limitations. In [2] and [16], the Lambertian radiation model is calibrated at the offline stage to make the RSS-distance relationship more consistent with the actual channel conditions. The RSS in [21] is converted into RSS ratio (RSSR) so that the priori information of the receiver height and transmission power in the channel model is not required. Additionally, intelligent optimization [11], [22] and machine learning [23], [24] techniques are utilized to improve localization accuracy. The work in [14] utilizes RSS data pre-processing and convolutional neural network (CNN) to mitigate light deficient regions in VLP systems. The sparse distribution of LEDs is considered in [10], [25], [26], where multiple PDs are usually deployed to compensate for the lack of transmitters.

However, to the best of our knowledge, the accumulation of dust on the surface of LEDs has not been considered in existing VLP literatures. In indoor areas, dust deposition is natural and inevitable, especially for the scenarios such as factories, warehouses, and underground garages. The dust particles may attenuate the signals and change the luminous characteristics of LEDs [27]. Consequently, in RSS VLP systems, the RSS-distance relationship obtained by the pre-existing channel model will gradually deviate from actual situations as dust accumulates, resulting in a continuous decline in localization performance. Dust accumulation is a slow and endless process. If we frequently calibrate the channel model of each LED based on a group of RSS measurements as in literatures (e.g., 12 and 29 measurements in [16] and [2], respectively), the system maintenance cost will be greatly increased and makes the VLP scheme impractical.

Therefore, in this work, we study the RSS VLP with dust accumulation taken into account. Our objective is to develop lightweight VLP methods to mitigate the impact of dust on the localization performance with reasonable measurement cost. To provide experimental support, we construct a testbed in a 3 m × 3 m × 3 m space with four LEDs and one PD. Our contributions are summarized as follows.

1) We first verify the impact of dust on the pre-existing channel model and the localization performance of RSS VLP system via extensive experiments under various dust accumulation scenarios. The results demonstrate that the positioning error is significantly increased even in slight dust accumulation cases, and centimeter-level positioning accuracy cannot be achieved as dust accumulates.
2) By introducing the correction factors α and β, we modify the classical Lambertian-based channel model to reflect the influence of dust. Then a two-points approach is proposed which performs the channel model calibration only based on two RSS measurements at pre-determined locations. The selection of optimal calibration locations is also discussed.
3) Based on substantial experimental results, the variations of the parameters in the modified channel model with the accumulation of dust are observed and mathematically modeled. Then a single-point approach is designed, which further reduces the channel model calibration cost to be a unique RSS measurement at the optimal calibration location.

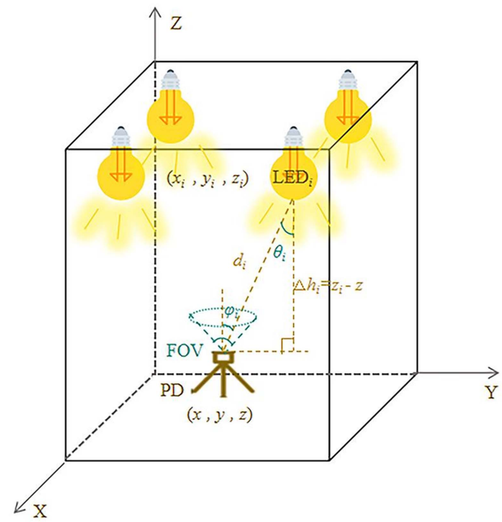

Fig. 1. Typical RSS-based indoor VLP system model.

The remainder of this paper is organized as follows. Section II introduces the system model and classical RSS VLP. Section III presents the impact of dust on the localization performance. The proposed scheme is described in Section IV, followed by performance evaluation in Section V. Finally, the paper is concluded in Section VI.

II. RSS-Based VLP System

A. Channel Model

A typical RSS-based indoor VLP system consists of *M* LED transmitters mounted on the ceiling and a PD receiver at the target position as shown in Fig. 1. The position coordinates of LEDs denoted by $(x_i,\ y_i,\ z_i)$ $(i = 1,\ 2, \ldots,\ M)$ are known, and the position of PD (*x*, *y*, *z*) is to be determined by VLP. Here, we mainly consider the LOS link, because more than 95% of the light intensity received by the receiver comes from the LOS component in most indoor scenarios and the non-line-of-sight (NLOS) component is usually negligible [6], [7], [28]. Let P_{t_i} represent the transmitted power of LED_i. Then the optical power received by PD can be expressed as

$$P_{r_i} = P_{t_i} H_i(0) \tag{1}$$

where $H_i(0)$ is the LOS channel gain between LED_i and the PD. Following the well-known Lambertian radiation law, we have [29], [30]

$$H_i(0) = \begin{cases} \frac{(m_i+1)A}{2\pi d_i^2} \cos^{m_i}(\theta_i) T_S(\phi_i) g(\phi_i) \cos(\phi_i), & 0 \le \phi_i \le \psi \\ 0, & \phi_i \ge \psi \end{cases} \tag{2}$$

where m_i is the Lambertian order of LED_i, *A* is the effective area of the PD, d_i is the distance between LED_i and the PD, θ_i, φ_i

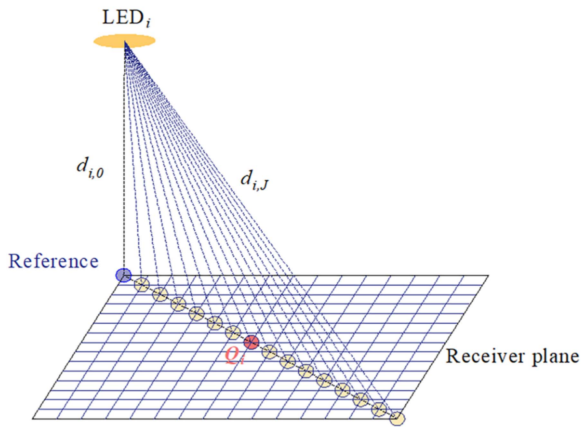

Fig. 2. Locations for the Lambertian order calibration.

represent the radiation and incident angles, respectively, and ψ is the field of view (FOV) of the PD. $T_S(\varphi_i)$ and $g(\varphi_i)$ denote the gains of optical filter and the concentrator, respectively, which are set to be 1 since these devices are used. Assuming that the transmitter and receiver planes are parallel and the relative vertical height of LED$_i$ to the PD is denoted by $\Delta h_i = z_i - z$, we have $\cos(\theta_i) = \cos(\varphi_i) = \Delta h_i / d_i$. Then (1) can be rewritten as

$$P_{r_i} = P_{t_i} \frac{(m_i + 1) A (\Delta h_i)^{m_i+1}}{2\pi d_i^{m_i+3}}. \tag{3}$$

B. Lambertian Order Calibration

Ideally, the parameters in the above channel model are priori known and are constant per LED and PD type. Moreover, the Lambertian order m_i is set to be 1 for a simple analysis in many literatures (e.g., [31], [32]). In practice however, the ideal channel model diverges from real-life measurements due to the variations of the LEDs' radiation pattern [33]. Therefore, some recent works calibrate the channel model (primarily the Lambertian order m_i), often even per LED, based on a group of offline measurements at known locations to fit the RSS-distance relation [2], [16], [33]. Nevertheless, how to determine the calibration locations is usually not discussed in literatures.

In order to achieve accurate positioning, we also calibrate the Lambertian order m_i of LEDs during the offline stage and provide a suggestion on the selection of calibration locations. At the receiver plane, the location directly underneath LED$_i$ is taken as the reference location because it normally presents the most information of the RSS-distance relationship [19]. At this location, the signal strength received by PD is denoted as $P_{r_{i,0}}$ and the distance to LED$_i$ is $d_{i,0}\ (= \Delta h_i)$. From the reference location, we expand outward along a straight line and measure the RSS at *J* locations for model calibration as illustrated in Fig. 2. The selected locations should cover all regions of the Lambertian propagation model including the plateau underneath the LED, steep descent in the middle, and the tail at the end [16]. At the *j*-th $(1 \leq j \leq J)$ calibration location, the received signal strength is denoted as $P_{r_{i,j}}$ and the distance to LED$_i$ is $d_{i,j}$. According to (3), we can obtain the ratio of the received signal strength at location *j* to the reference location as follows:

$$\frac{P_{r_{i,j}}}{P_{r_{i,0}}} = \left(\frac{d_{i,0}}{d_{i,j}}\right)^{m_i+3}. \tag{4}$$

Then the Lambertian order m_i can be calibrated based on the RSS measurements at *J* locations, which is

$$m_i = \frac{1}{J}\left(\sum_{j=1}^{J}\left(\frac{\ln\left(\frac{P_{r_{i,j}}}{P_{r_{i,0}}}\right)}{\ln\left(\frac{d_{i,0}}{d_{i,j}}\right)} - 3\right)\right). \tag{5}$$

C. The Optimal Calibration Location

The Lambertian order m_i obtained by (5) is the average of *J* RSS measurements. If only one location is selected for calibration, e.g., location *j*, the calibrated Lambertian order is denoted by $m_{i,j}$ as follows:

$$m_{i,j} = \frac{\ln\left(\frac{P_{r_{i,j}}}{P_{r_{i,0}}}\right)}{\ln\left(\frac{d_{i,0}}{d_{i,j}}\right)} - 3. \tag{6}$$

Among *J* locations, we select the one with $m_{i,j}$ closest to the averaged m_i and define it as the optimal calibration location Q_i as shown in Fig. 2, that is

$$Q_i = \arg\min_j |m_i - m_{i,j}|. \tag{7}$$

This means the calibration effect of the single location Q_i is similar to the average of *J* locations. We will measure the RSS at location Q_i and the reference location for a fast calibration in dust deposition scenarios as discussed in Section IV.

D. Online Positioning

Once the channel model is calibrated during the offline stage and the RSS-distance relationship is built for each LED, the PD at the target position can measure RSS values from available LEDs and then calculate the corresponding distance as follows:

$$d_i = \left(\frac{P_{t_i}(m_i + 1) A (\Delta h_i)^{m_i+1}}{2\pi P_{r_i}}\right)^{\frac{1}{m_i+3}}. \tag{8}$$

Furthermore, the least square estimation (LSE) is deployed to determine the coordinates of the target position [3].

III. The Impact of Dust Accumulation on Positioning

In indoor areas, dust deposition is natural and inevitable, especially for industrial environments. It has been discussed in [27] that the dust particles may attenuate the signals and change the luminous characteristics of LEDs in visible light communication applications, however, their influence to indoor positioning is still unclear. Therefore, in this section, we verify the impact of dust on the localization performance of RSS VLP through extensive experiments. We deploy the RSS VLP scheme described in Section II, which calibrates the channel model based on 12 offline RSS measurements as in [16]. The detailed testbed settings are provided in Section V.

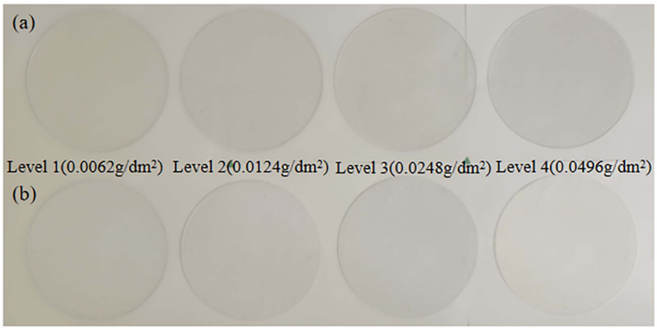

Fig. 3. LED panels with different densities of dust. (a) Dark dust. (b) White dust.

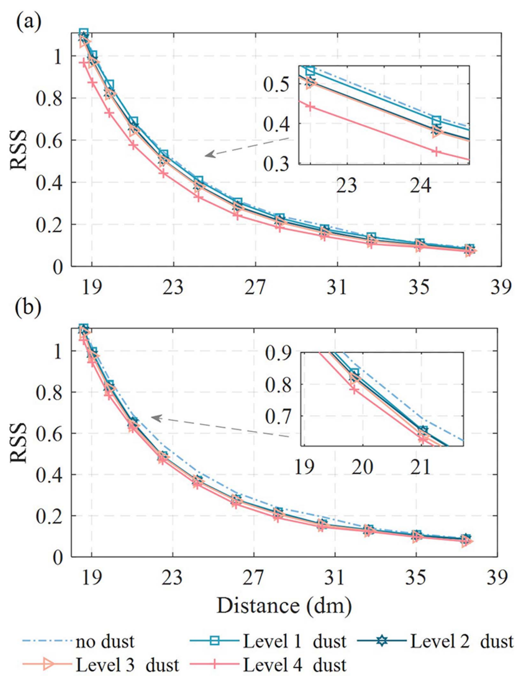

Fig. 4. Comparison of the fitted RSS-Distance curve with actual RSS values measured in different dust accumulation scenarios. (a) Dark dust. (b) White dust.

It is known that the dust composition is complex and affected by various factors such as outdoor environment, indoor activities, facility operation, and building materials [34], [35]. In the experiment, we examined two types of dust with different compositions to simulate real-life dust accumulation. One type mainly consists of a mixture of soil and appears a darker color (referred to as dark dust). The other type mainly consists of SiO_2 and appears a lighter color (referred to as white dust). For each type of dust, four degrees of concentration were tested. The dust densities are $0.0062\,\mathrm{g/dm^2}$ (Level 1), $0.0124\,\mathrm{g/dm^2}$ (Level 2), $0.0248\,\mathrm{g/dm^2}$ (Level 3) and $0.0496\,\mathrm{g/dm^2}$ (Level 4), respectively. The actual LED panels with evenly distributed dust are shown in Fig. 3.

Fig. 4 compares the RSS-distance curve (the blue dashed line) fitted based on the channel model of LED_1 with the actual RSS values measured in different dust accumulation scenarios. It can be observed that the received signal strength at the same location gradually decreases with the accumulation of dust as

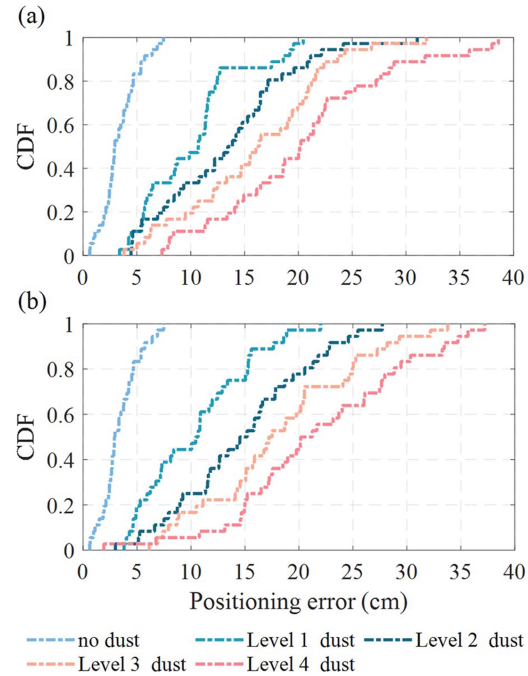

Fig. 5. The CDF of positioning error. (a) Dark dust. (b) White dust.

expected, and the initially fitted RSS-distance curve increasingly diverges from actual values. In other words, the channel model established at the offline stage constantly deviates from actual situations and loses its correctness as dust accumulates. The consequence of this model deviation on the positioning performance is clearly demonstrated by Fig. 5, where the cumulative distribution functions (CDFs) of positioning errors in different dust accumulation scenarios are shown. We can see that two types of dust have similar effects on the localization performance. Initially, when there is no dust, 80% of the positioning error is less than 5 cm and the maximum positioning error is less than 8 cm. However, in the slightest dust accumulation scenario (Level 1), 80% of the positioning error approaches 15 cm and the maximum positioning error exceeds 20 cm for both types of dust. When the dust accumulation is heavy (Level 4), the maximum error even approaches 40 cm.

Extensive experimental results demonstrate that dust deposition is crucial for the RSS VLP scheme and cannot be ignored. Even for the slightly visible dust cases, the significantly increased positioning error is still unacceptable for VLP with centimeter-level positioning accuracy.

IV. The Proposed RSS VLP Scheme Considering Dust Accumulation

As discussed above, although the channel model is calibrated at the offline stage to adapt to the LED's radiation pattern, it still gradually deviates from actual situations as dust accumulates and introduces significant positioning errors as a result. Dust deposition is a slow and endless process. If we periodically recalibrate the channel model of each LED based on a group of RSS measurements as in the initial offline stage (e.g., 12

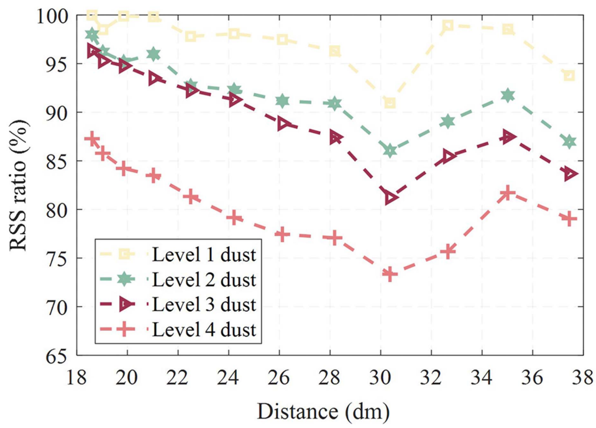

Fig. 6. The ratio of the RSSs measured in dust accumulation scenarios to the ones in the initial dust-free scenario.

and 29 measurements in [16] and [2], respectively), the maintenance cost will be greatly increased and makes the VLP system impractical. Therefore, in this section, we first investigate the channel model with dust accumulation taken into account, and then propose two lightweight calibration methods for mitigating the dust impact and improving the positioning accuracy.

A. Channel Model Considering Dust Accumulation

It can be seen from (3) that there are two parameters related to the transmitter in the channel model, i.e., P_{t_i} and m_i. If dust only affects the transmitted power P_{t_i}, the power received at different locations should decrease with the same proportion. Taking the results of LED$_1$ as an example, the ratios of the RSS measured in dust accumulation scenarios to the ones in the initial dust-free scenario are shown in Fig. 6. We can observe that the RSS ratio curves seem irregular, that is, the power received at different locations does not decrease with equal proportion. This indicates that the dust accumulation not only reduces the transmitted power but also changes the radiation pattern of LEDs, i.e., both P_{t_i} and m_i are changed by dust.

Based on the above analysis, we define the received power, the transmitted power and the Lambertian order of LED$_i$ in the case of dust accumulation as $\hat{P}_{r_i}$, $\hat{P}_{t_i}$ and $\hat{m}_i$, respectively. Then the relationship between the received power and the distance can be modified as:

$$\hat{P}_{r_i} = \hat{P}_{t_i} \frac{(\hat{m}_i + 1)A(\Delta h_i)^{\hat{m}_i+1}}{2\pi d_i^{\hat{m}_i+3}}. \tag{9}$$

B. Two-Points Fast Calibration

At the initial offline stage, the channel model is calibrated according to (5) based on the RSSs measured at the reference location and J calibration locations. Also, the location Q_i is selected out from J locations. The calibration effect of location Q_i is similar to the average of J measurements. This implies that most of the calibration information is contained in the reference location and location Q_i.

Therefore, we propose a two-points approach to calibrate the channel model of (9) in dust accumulation scenarios, which requires only two RSS measurements at the reference location and location Q_i. We denote the RSSs measured at these two locations as $\hat{P}_{r_{i,0}}$ and $\hat{P}_{r_{i,Q}}$, respectively. According to (4), the ratio of the RSS received at location Q_i to that of the reference location can be written as

$$\frac{\hat{P}_{r_{i,Q}}}{\hat{P}_{r_{i,0}}} = \left(\frac{d_{i,0}}{d_{i,Q}}\right)^{\hat{m}_i+3}. \tag{10}$$

By combining (4) and (10), the Lambertian order of LED$_i$ in the case of dust accumulation can be given by

$$\hat{m}_i = m_i - \frac{\ln\left(\frac{P_{r_{i,Q}}\hat{P}_{r_{i,0}}}{P_{r_{i,0}}\hat{P}_{r_{i,Q}}}\right)}{\ln\left(\frac{d_{i,0}}{d_{i,Q}}\right)}. \tag{11}$$

For the convenience of the following description, (11) is simplified as

$$\hat{m}_i = m_i - \beta \tag{12}$$

where β can be regarded as a correction factor induced by dust.

Then the relationship between the received signal strength and the distance at location Q_i can be modeled as

$$\hat{P}_{r_{i,Q}} = \hat{P}_{t_i} \frac{(m_i - \beta + 1)A(d_{i,0})^{m_i-\beta+1}}{2\pi d_{i,Q}^{m_i-\beta+3}}. \tag{13}$$

By comparing (13) with the RSS measured at location Q_i during the initial offline stage (i.e.,$P_{r_{i,Q}}$), the transmit power of LED$_i$ in the case of dust accumulation can be calibrated as follows:

$$\hat{P}_{t_i} = \frac{\hat{P}_{r_{i,Q}}}{P_{r_{i,Q}}} \frac{(m_i + 1)}{(m_i - \beta + 1)} \left(\frac{d_{i,0}}{d_{i,Q}}\right)^{\beta} P_{t_i}. \tag{14}$$

In the same way, we simplify (14) as

$$\hat{P}_{t_i} = \alpha P_{t_i} \tag{15}$$

where α can also be regarded as a correction factor for dust.

In summary, the channel model calibrated by the proposed two-points approach can be expressed as

$$\hat{P}_{r_i} = \alpha P_{t_i} \frac{(m_i - \beta + 1)A(d_{i,0})^{m_i-\beta+1}}{2\pi d_i^{m_i-\beta+3}}. \tag{16}$$

The correction factors α and β can be calculated as follows:

$$\begin{cases} \alpha = \frac{\hat{P}_{r_{i,Q}}}{P_{r_{i,Q}}} \frac{(m_i+1)}{(m_i-\beta+1)} \left(\frac{d_{i,0}}{d_{i,Q}}\right)^{\beta} \\ \beta = \frac{\ln\left(\frac{P_{r_{i,Q}}\hat{P}_{r_{i,0}}}{P_{r_{i,0}}\hat{P}_{r_{i,Q}}}\right)}{\ln\left(\frac{d_{i,0}}{d_{i,Q}}\right)}. \end{cases} \tag{17}$$

C. Single-Point Fast Calibration

In the experiments, we observed the variations of the parameters in (16) and (17) with the accumulation of dust. It was found that there are approximately linear inverse relationships between the dust density and the channel model correction factors α, β as well as the RSS measured at location Q_i (i.e., $\hat{P}_{r_{i,Q}}$) as shown in

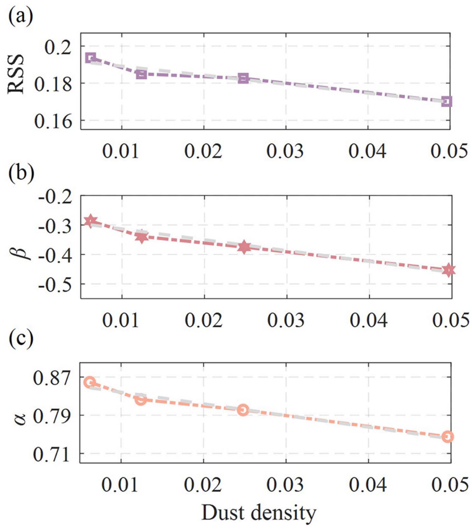

Fig. 7. The relationships between the parameters in the calibrated channel model and the dust density.

Fig. 7. Based on this finding, we further propose a single-point model calibration approach, which only requires a unique RSS measurement at location Q_i.

Here, we mathematically model the relationship curves shown in Fig. 7 as follows

$$\begin{cases} \rho = a_1 \hat{P}_{r_{i,Q}} + b_1 \\ \alpha = a_2 \rho + b_2 \\ \beta = a_3 \rho + b_3 \end{cases} \tag{18}$$

where the coefficients a_1, a_2, a_3, b_1, b_2 and b_3 can be obtained by the linear fitting of the relationship curves in Fig. 7. Once the RSS at location Q_i is measured by PD, the model correction factors α and β can be calculated according to (18) instead of (17), and then the channel model can be calibrated by (16). This means the measurement at the reference location is no longer required. It should be mentioned that a prerequisite to this single-point approach is the pre-built relationship curves in Fig. 7.

D. Proposed RSS VLP Scheme

In summary, the proposed RSS VLP scheme includes three stages: initial offline stage, periodic calibration stage and online positioning stage. At the initial offline stage, the PD measures the RSS values at the underneath reference location and J selected locations to calibrate the Lambertian order m_i of each LED and establish the channel model as described in Section II. At the end of this stage, the PD finds out location Q_i from J selected locations for the following periodic calibration.

Thereafter, the PD periodically measures the RSSs at the reference location as well as location Q_i, and then updates the channel model based on (16) and (17) by the proposed two-points calibration approach. The update period depends on the speed of dust accumulation. During this periodic calibration stage, the PD continuously collects the parameters including α, β and the RSS value measured at location Q_i. Once the relationship curves between these parameters and the dust density are established as in Fig. 7, we can change to use the proposed single-point approach for a more lightweight calibration, by which the channel model can be quickly calibrated only based on a unique RSS measurement at location Q_i according to (16) and (18).

The online positioning stage starts just after the initial offline stage. At the target position, the PD measures the RSS values of available LEDs and then calculates their corresponding distances according to (8) based on the latest calibrated channel model. The target coordinates can be further estimated by least square method.

By our proposed RSS VLP scheme, in order to improve the positioning accuracy, the channel model is calibrated not only in the initial offline stage to adapt to the LEDs' radiation pattern, but also periodically calibrated afterwards to mitigate the impact of dust. But each periodic calibration requires only two or even one RSS measurement. The joint consideration on the positioning performance and the maintenance cost ensures the implementability of the system. It is worthy to be mentioned that the channel model calibrations of LEDs are independent to each other, which means our proposed methods can be used for a single LED.

V. Performance Evaluation

In this section, we conduct experimental research to evaluate the accuracy of the proposed channel calibration approaches and the localization performance of the RSS VLP scheme. We first introduce the testbed settings and the key parameters adopted, and then analyze the experimental results under different dust accumulation scenarios.

A. Testbed Settings

An indoor VLP system is built in a 3 m $\times$ 3 m $\times$ 3 m space as shown in Fig. 8, where 4 off-the-shelf LEDs are installed at the corners with coordinates of (2.6, 2.6, 2.65), (2.6, 0.4, 2.65), (0.4, 0.4, 2.65) and (0.4, 0.4, 2.65) in meters, respectively. Fig. 8(a) presents the implementation of the LED transmitter, where square-wave signals of different frequencies are modulated by microcontrollers and subsequently amplified by linear power amplifiers (OPA541) to drive the LEDs. The signals from LEDs are distinguished by the frequencies with a requirement of avoiding the superposition of their harmonic components in frequency domain and the visible flicker of lights [36]. Based on these considerations, the signal frequencies for 4 LEDs are set as 22.5 kHz, 32 kHz, 62.5 kHz, and 72 kHz, respectively.

At the receiver side, the PD (Thorlabs FDS1010) is placed horizontally upwards at 0.8 m above the ground. The PD receives the mixed optical signal from the LEDs and converts it into a current signal, which is further converted into a voltage signal and amplified through the current-voltage conversion and amplification circuits (LT1818IS8). The microcontroller samples the amplified signal at a rate of 512 kHz and then performs the fast Fourier transform (FFT) in groups of 1024 sampling points. The 1st harmonic of each signal in frequency domain is extracted as the RSS value of the corresponding LED, since which is linearly proportional to the amplitude of the square wave in time

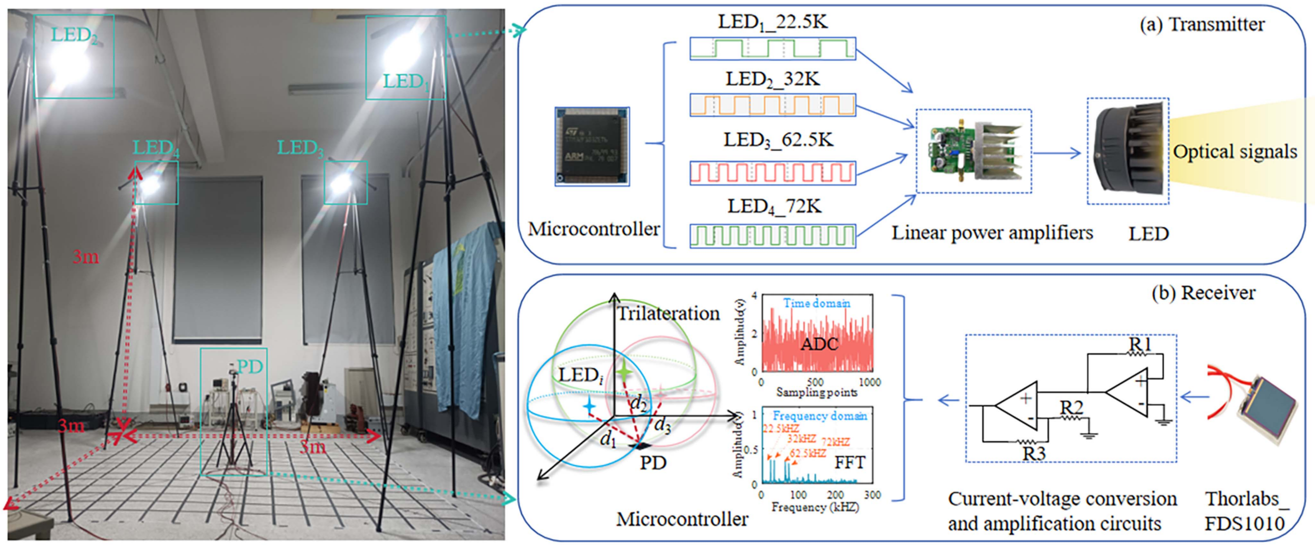

Fig. 8. Testbed environment. (a) Components of the transmitter. (b) Components of the receiver.

TABLE I
THE CALIBRATED LAMBERTIAN ORDER OF LEDS

LED_i $(i = 1, 2, 3, 4)$	m_i $(i = 1, 2, 3, 4)$
LED_1	0.710
LED_2	0.742
LED_3	0.516
LED_4	0.733

domain [24], [37]. FFT operation is a natural bandpass filter for noises (e.g., thermal noise). Moreover, we take the average of 20 measurements at each position as the final RSS value to avoid sampling noise. The positioning area is divided into small squares of 0.2 m $\times$ 0.2 m, and the locations at an interval of 0.8 m in both the horizontal and vertical directions selected for observation.

B. The Effect of Lambertian Order Calibration

At the initial offline stage, the Lambertian order m_i is calibrated by our proposed scheme to adapt to the LEDs' real radiation pattern as discussed in Section II-B. The RSS values are measured at the underneath reference location and $J = 11$ calibration locations that are evenly distributed on the diagonal of LEDs. Therefore, 12 RSS measurements are needed for each channel model calibration as in [16]. The calibrated m_i of LEDs are listed in Table I.

Fig. 9 shows a comparison of the positioning results with m_i calibrated and uncalibrated (i.e., $m_i = 1$ as in [31], [32]). The experiment was conducted after the initial offline stage. It can be seen from Fig. 9(a) that the RSS values of LED_1 estimated by the calibrated channel model are closer to the actual ones.

Consequently, a much better localization performance is achieved as shown in Fig. 9(b), where the 80% positioning error is reduced from 11 cm to 4.7 cm by the channel model calibration.

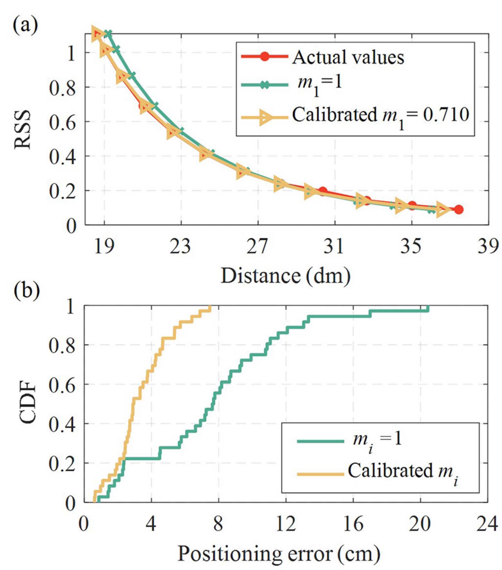

Fig. 9. The necessity of calibrating the channel model at the initial offline stage. (a) Comparison of actual and estimated RSS values; (b) Comparison of positioning accuracy.

C. The Performance of the Proposed RSS VLP Scheme

To evaluate the performance of the proposed VLP scheme, extensive experiments were conducted under various dust accumulation scenarios. As discussed in Section III, the impacts of dust with different compositions on the system performance are quite similar; hence, this section mainly presents the experimental results obtained in dark dust scenarios. Since dust accumulation has not been considered in related works to the best of our knowledge, the proposed scheme is compared with

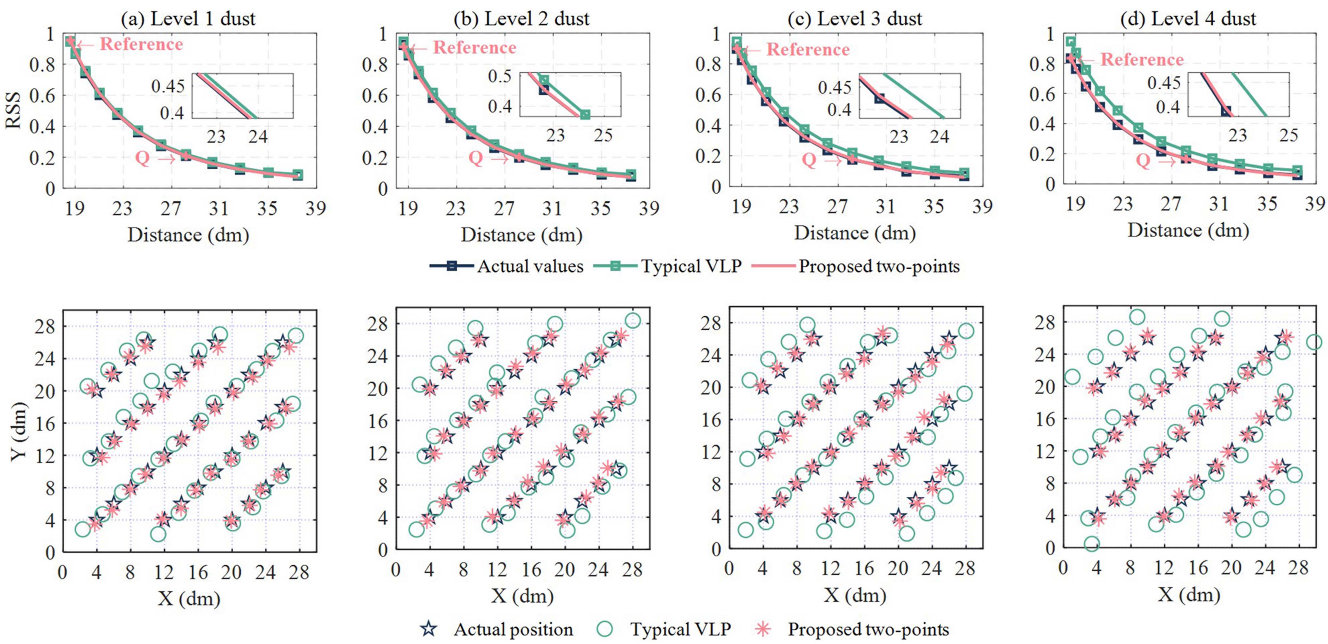

Fig. 10. Comparisons of the estimated RSSs and positions with actual ones under different dust accumulation scenarios.

the scheme presented in [16] (referred to as Typical VLP), which is also model-based RSS VLP and calibrates the Lambertian order m_i based on 12 offline measurements.

1) Two-Points Calibration Approach: By our proposed two-points approach, the channel model is periodically calibrated according to (16) and (17), based on two RSSs measured at the reference location and location Q_i. Location Q_i is determined at the initial offline stage according to (7). The first row of Fig. 10 compares the RSS values estimated by two schemes with the actually measured results under different dust accumulation scenarios, and the corresponding positioning results are intuitively compared in the second row. It can be found from Fig. 10 that the channel model calibrated by our two-points approach matches the actual results well, even for the heavy dust deposition scenarios. The estimated positions almost overlap the actual positions. These results indirectly demonstrate the significant calibration effect of location Q_i. Because the Typical VLP scheme calibrates the channel model only at the initial offline stage, the estimated RSSs increasingly diverge from actual values as dust accumulates, and obvious deviations from actual positions can be observed especially at the edge of positioning area. However, if this scheme also periodically calibrates the channel model, the greatly increased maintenance cost caused by 12 RSS measurements per LED for each calibration will make the system impractical.

Figs. 11 and 12 provide the statistical results of positioning error in terms of CDF and average, respectively. We can see from Fig. 11 that by our proposed scheme, the maximum positioning error remains below 10 cm and 80% of the error is within 6 cm. The impact of dust accumulation is not significant. However, the positioning accuracy of Typical VLP scheme obviously decreases as dust accumulates and the maximum positioning

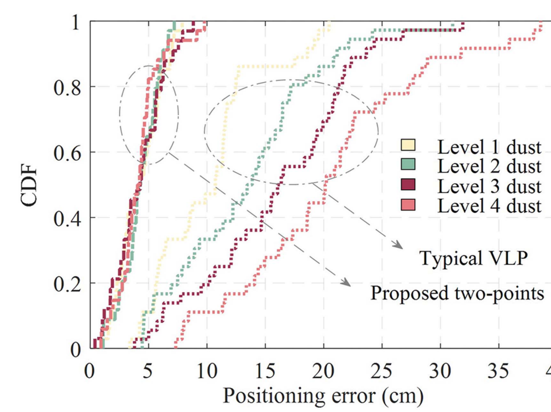

Fig. 11. The CDF of positioning error under different dust accumulation scenarios.

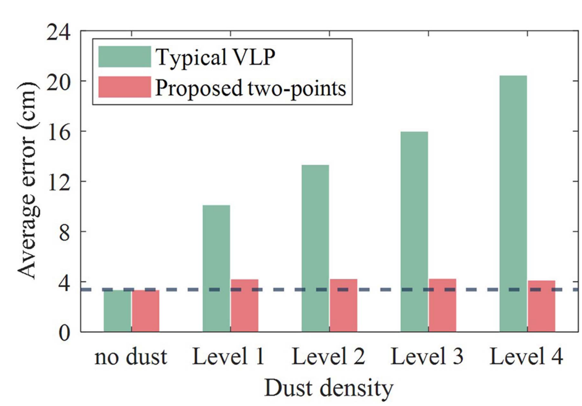

Fig. 12. The average positioning error under different dust accumulation scenarios.

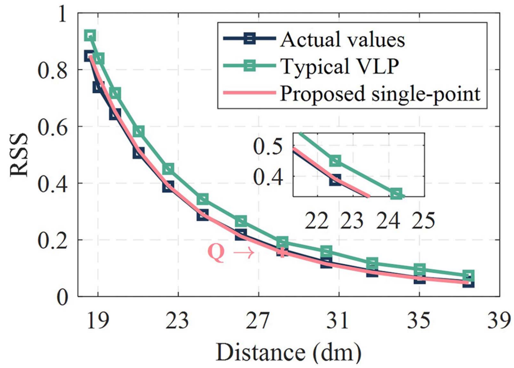

Fig. 13. The comparison of estimated RSS values with actually measured values.

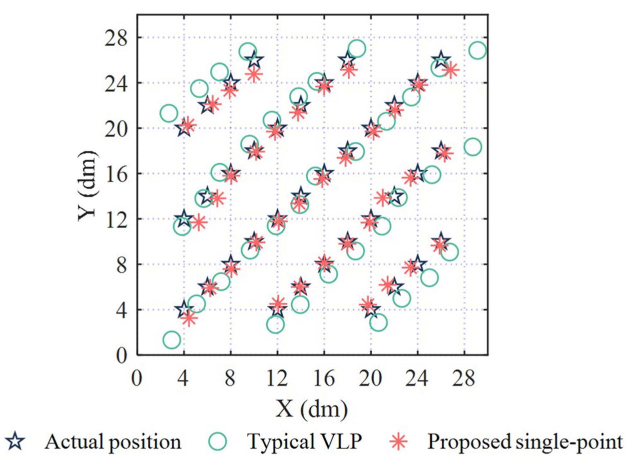

Fig. 14. The comparison of predicted locations with actual locations.

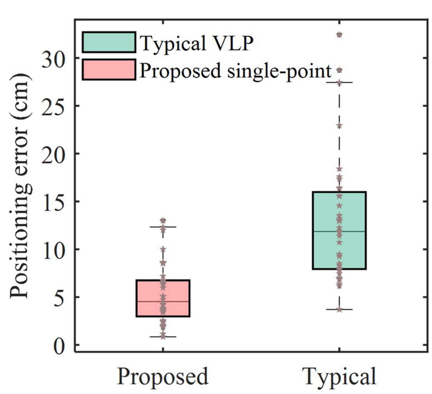

Fig. 15. The comparison of predicted locations with actual locations.

error even approaches 40 cm. It is shown in Fig. 12 that the average error of the proposed scheme always remains around 4 cm, which is very close to that in the initial dust-free scenario (3.4 cm). The impact of dust on the positioning accuracy can be effectively mitigated by our two-points calibration approach. On the other hand, the experimental results also demonstrate that if the Lambertian model is properly calibrated, it can accurately estimate the channel conditions in various dust accumulation scenarios, thereby ensuring the positioning accuracy of RSS VLP.

2) Single-Point Fast Calibration Approach: As described in Section IV-C, after the VLP system runs for a period of time, enough RSS information of location Q_i can be obtained from the periodic two-points calibration process. Then the relationships between the dust density and the RSS as well as the model correction factors α, β can be established as in Fig. 7. Thereafter, we change to use single-point calibration approach to further reduce the measurement cost. The channel model is calibrated only based on one RSS measured at location Q_i.

We evaluate the performance of the single-point approach in a new dust accumulation scenario where the curves in Fig. 7 have been generated. It is shown in Fig. 13 that the RSS values estimated by our single-point approach are very close the actually measured ones. That is, the channel model rebuilt by our single-point approach based on one RSS also fits the actual situation well. Consequently, a better positioning accuracy can be achieved compared to Typical VLP as illustrated in Fig. 14. The positioning errors obtained by two schemes are statistically compared by the boxplots in Fig. 15.

It is noticeable that a few results of the proposed single-point approach exceed 10 cm while the mean error is still around 5 cm. Compared to Typical VLP, a higher positioning precision can be achieved by the proposed scheme.

VI. Conclusion

In this paper, a lightweight RSS VLP scheme is presented jointly considering dust accumulation and system maintenance cost. The nonnegligible negative impact of dust on the localization performance of RSS VLP is firstly confirmed through experiments, which reals that the positioning error is greatly increased in dust accumulation scenarios due to the mismatch of the pre-built channel model. Therefore, we first modify the classical Lambertian-based channel model by introducing the correction factors α and β to reflect the influence of dust, and then propose a two-points approach to calibrate the channel model based on two RSS measurements at the reference location and pre-determined location Q_i. Furthermore, the relationships between the channel model parameters and the dust density are mathematically modeled, and a single-point approach is designed which calibrates the channel model only based on the RSS measured at location Q_i. Experimental results demonstrate that the average positioning error can be maintained below 5 cm by the proposed two-points approach and 6 cm by the single-point approach under different dust accumulation scenarios. The impact of dust can be effectively mitigated by the proposed RSS VLP scheme. Moreover, each periodic channel model calibration performed by our approaches only requires two or even one RSS measurement, which further ensures the system feasibility and implementability.

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